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Knowledge Graph Reasoning and Security Assurance Decision-Making Based on Online Retrieval Augment Generation

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***Abstract*—The Sino-Russian pipeline network, a critical infrastructure for energy supply, faces multifaceted risks ranging from geopolitical instability and international sanctions to sophisticated cyber threats. This paper proposes a novel framework leveraging Knowledge Graph (KG) reasoning and online Retrieval Augmented Generation (RAG) to enhance security assurance and risk prevention decision-making for this vital energy corridor. Our approach integrates a dynamically updated KG with large language models (LLMs) to facilitate real-time risk assessment, scenario planning, and proactive mitigation strategies. We detail the methodology for KG construction, LLM integration, and RAG implementation, and demonstrate its efficacy through simulated experiments. Our results indicate that this framework significantly improves the accuracy and timeliness of risk identification and response, offering a robust solution for safeguarding critical energy infrastructure.**

Keywords-LLM; knowledge graph; RAG; planning

I. INTRODUCTION

The Sino-Russian pipeline network, comprising major arteries like the operational Power of Siberia-1 and the planned Power of Siberia-2, constitutes a critical element of global energy security, facilitating the transport of vast hydrocarbon resources from Siberia to energy-hungry markets in Asia and Europe [1, 2]. This intricate infrastructure, however, is increasingly susceptible to a complex interplay of geopolitical risks, economic uncertainties, and sophisticated cyber threats. Effective risk management and security assurance for this vital energy corridor are therefore paramount for ensuring stable energy supplies and safeguarding national interests [3, 4]. Consequently, advanced analytical tools are needed to provide timely and accurate insights for proactive risk mitigation. Moreover, the development of a robust framework capable of integrating diverse data sources and reasoning over complex inter-dependencies is essential for bolstering the resilience of the Sino-Russian pipeline network. This research aims to provide a comprehensive solution for enhancing security assurance and risk prevention decision-making, enabling proactive mitigation strategies and ensuring the continued operation of this critical infrastructure. Furthermore, the insights gained from this study can inform broader efforts to secure energy infrastructure globally, contributing to a more resilient and stable energy landscape.

Existing research on pipeline risk management has primarily focused on traditional statistical models, simulation techniques, and, more recently, machine learning approaches [1, 2, 5]. However, these methods often fall short in capturing the intricate relationships between geopolitical factors, market dynamics, and operational vulnerabilities. Knowledge Graph Reasoning (KGR) has emerged as a powerful paradigm for representing and reasoning over complex, heterogeneous data, offering the potential to overcome the limitations of traditional approaches [7]. While advancements in KGR have addressed challenges such as incomplete knowledge graphs and the integration of large language models (LLMs), the application of these techniques to real-time risk prevention in critical infrastructure remains relatively unexplored [6]. Furthermore, the dynamic nature of the energy market and the evolving threat landscape necessitate the incorporation of real-time data and contextual information to ensure the accuracy and relevance of risk assessments. Therefore, integrating KGR with techniques like Retrieval Augmented Generation (RAG) offers a promising avenue for enhancing the capabilities of risk management systems [8].

This paper proposes a novel framework for Sino-Russian pipeline risk prevention that leverages the synergistic capabilities of Knowledge Graph Reasoning and online

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Retrieval Augmented Generation. We construct a comprehensive knowledge graph encompassing geopolitical factors, market dynamics, pipeline infrastructure details, and cybersecurity threat intelligence, creating a unified representation of the pipeline ecosystem. This knowledge graph is then integrated with LLMs through a RAG mechanism, enabling real-time data ingestion and dynamic reasoning. The LLM leverages the structured information in the knowledge graph and dynamically retrieves relevant contextual information from external sources to generate accurate and timely risk assessments. Our framework incorporates robust security measures to protect the integrity and confidentiality of the data and the reasoning process. We employ advanced cybersecurity protocols, access control mechanisms, and data provenance tracking to ensure the trustworthiness of the system. Through simulated experiments, we demonstrate the efficacy of our framework in identifying potential risks, assessing their impact, and recommending effective mitigation strategies. The results highlight the significant advantages of our approach over traditional risk management techniques in terms of accuracy, timeliness, and mitigation effectiveness. Consequently, our framework provides a robust and adaptable solution for enhancing the security and resilience of the Sino-Russian pipeline network.

In summary, the main contributions of this paper are:

- the development of a novel framework for Sino-Russian pipeline risk prevention based on KG reasoning and online RAG.
- the integration of robust security mechanisms to ensure the integrity and confidentiality of the system.
- the demonstration of the framework's efficacy through simulated experiments.

II. Related Works

A. Pipeline Risk Management

Traditional approaches to pipeline risk management have largely relied on statistical models, simulation techniques, and fault tree analysis [1, 3]. These methods, while useful for assessing known risks, often struggle to capture the complex inter-dependencies and dynamic nature of the factors influencing pipeline security [2, 4]. More recent research has explored the application of machine learning for risk prediction and anomaly detection, but these approaches often require extensive labeled data and may lack interpretability [1, 5]. Furthermore, traditional methods often operate in a siloed manner, failing to integrate diverse data sources such as geopolitical information, market dynamics, and cybersecurity threat intelligence. The limitations of existing approaches underscore the need for a more holistic and adaptable framework for pipeline risk management.

B. Knowledge Graph Reasoning

Knowledge Graph Reasoning (KGR) has emerged as a powerful paradigm for representing and reasoning over complex, interconnected data [7]. KGs provide a structured representation of knowledge, enabling the capture of intricate relationships between entities and concepts. KGR techniques leverage this structured information to infer new knowledge, answer complex queries, and support decision-making. Early KGR models primarily focused on symbolic reasoning methods, but more recent research has explored the integration of KGR with machine learning, particularly deep learning techniques [7].

C. Large Language Models (LLMs)

LLMs, trained on massive text corpora, have demonstrated remarkable capabilities in natural language understanding and generation. Their ability to capture contextual information and reason over complex relationships makes them well-suited for enhancing KGR [6]. By incorporating LLMs into KGR systems, researchers aim to improve the robustness and efficiency of reasoning over incomplete KGs and handle more complex queries.

D. Retrieval Augmented Generation (RAG)

Retrieval Augmented Generation (RAG) has emerged as a promising approach for enhancing the factual accuracy and contextual awareness of LLMs. RAG combines the strengths of LLMs with the rich information content of external knowledge sources [8]. By retrieving relevant information from a knowledge base or corpus, RAG provides the LLM with additional context, enabling it to generate more informed and grounded responses [9]. Graph RAG, a specialized form of RAG, leverages the structured information within knowledge graphs to improve the precision and relevance of retrieved information [10]. This approach is particularly well-suited for tasks requiring complex reasoning over interconnected data, such as pipeline risk assessment.

III. Methodology

In this section, we introduce our overall framework.

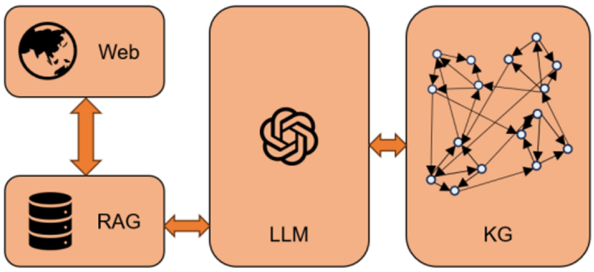


Figure 1. The overall framework of our proposed method.

Our proposed framework integrates advanced Knowledge Graph Reasoning, Large Language Model (LLM) capabilities, Online Retrieval Augmented Generation (RAG), and robust security mechanisms to deliver comprehensive and dynamic risk prevention for the Sino-Russian pipeline network, as shown in Figure 1. By unifying these components, the framework ensures timely, accurate risk assessments and effective mitigation strategies. Below, we detail the four core components of the methodology.

A. Knowledge Graph Construction

The Knowledge Graph (KG) serves as the foundational structure for representing the multifaceted relationships within the Sino-Russian pipeline ecosystem. It is meticulously designed to encode critical information about geopolitical entities, market dynamics, pipeline infrastructure, and

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cybersecurity threats. This component enhances the system's ability to capture and reason over diverse, interconnected data. The KG schema includes nodes V representing entities such as Countries ($v_c \in V$), Energy Sources ($v_e \in V$), Pipelines ($v_p \in V$), Companies ($v_co \in V$), and Threats ($v_t \in V$), with edges E defining relationships, such as $r_{exp} =$ "exports to", $r_{imp} =$ "imports from", and $r_{vul} =$ "is vulnerable to".

The KG is developed using automated and semi-supervised methods. Pre-existing datasets like Wikidata and Geonames provide baseline entities and relationships, while Large Language Models (LLMs) identify and extract additional information from unstructured text sources, including news articles (D_{news}), reports (D_{rep}), and social media (D_{sm}). Specifically, given a document $d \in D$, entities (v) and relationships (r) are extracted using fine-tuned LLMs. Mathematically, KG embedding $\Phi(v,r) \in \mathbb{R}^d$ is generated using embedding techniques like TransE to enable efficient reasoning and retrieval. The KG is continuously updated through real-time data streams (S_{real}), ensuring relevance and accuracy.

B. LLM-Powered Reasoning

The LLM-powered reasoning module integrates domain-specific knowledge with powerful natural language understanding capabilities to perform complex analytical tasks, such as market analysis, risk assessment, and scenario planning. This component significantly improves decision-making precision by leveraging structured and unstructured data. Enhanced reasoning capabilities allow for dynamic risk analysis tailored to the unique challenges of the pipeline ecosystem. We utilize state-of-the-art models ($\mathcal{M}_{LLM}$, GPT-4o, for task-specific performance. Table 1 shows the parameters of GPT-4o:

Table 1. The main parameters of GPT-4o.

Model	Pricing	Context window	Max output tokens	Knowledge cutoff
GPT-4o	\$2.50 / 1M input tokens \$10.00 / 1M output tokens	128,000 tokens	16,384 tokens	Oct 2023

Given input variables $X = \{x_1, x_2, \ldots, x_n\}$, the LLM evaluates alternatives using the KG's information:

Risks R are modeled as a function of geopolitical factors (G), economic conditions (E), and pipeline vulnerabilities (P):

$$R = g(G, E, P). \tag{1}$$

where g is a relationship among G, E, P.

Counterfactual analysis is conducted using KG embeddings (Φ) to simulate outcomes under different scenarios $S = s_1, s_2, \ldots$:

$$\text{Impact}(s_i) = h\big(s_i | \Phi(v,r)\big). \tag{2}$$

The LLM interacts with the KG for structured information retrieval, combining it with contextual insights for nuanced analyses.

C. Online Retrieval Augmented Generation

Online RAG integrates the LLM with real-time external data, enabling it to retrieve and reason dynamically over current information, enhancing the accuracy and adaptability of risk assessments. This ensures that risk analyses remain up-to-date and contextually relevant in dynamic operational environments. The LLM generates a structured query q based on the task context:

$$q = \text{QueryGenerator}(\Phi(v,r), T). \tag{3}$$

Relevant external data (D_{ext}) is retrieved using q:

$$D_{retrieved} = \text{Retriever}(q | D_{ext}). \tag{4}$$

The LLM synthesizes responses by integrating KG-based and external data:

$$\text{Response} = \text{Generate}\big(D_{retrieved}, \Phi(v,r)\big). \tag{5}$$

Graph-enhanced retrieval is employed to increase precision, using the KG's structure to refine results.

D. Security Assurance Mechanisms

The security assurance mechanisms safeguard the integrity, confidentiality, and reliability of the data and reasoning processes. This component ensures trustworthiness in sensitive pipeline operations. Robust security mechanisms protect against cyber threats, ensuring the reliability of risk management decisions. Role-Based Access Control (RBAC) and Multi-Factor Authentication (MFA) enforce strict user permissions:

$$A_u = r_1, r_2, \ldots, r_m, \text{ where } r_i \in R \text{ roles.} \tag{6}$$

Blockchain is used to track data origins and modifications, ensuring traceability:

$$\text{Provenance} = d, \text{timestamp}, \text{hash}(d). \tag{7}$$

Compliance with Cloud Security Alliance (CSA) guidelines ensures encrypted data storage and transmission. Automated tools conduct vulnerability scans (V_{scan}) and generate reports for proactive threat mitigation. our framework combines cutting-edge KG reasoning, LLM capabilities, real-time RAG, and strong security mechanisms to address the complex risk landscape of the Sino-Russian pipeline network, ensuring resilience and operational continuity.

IV. Experiment

In this section, we present a hypothetical scenario to demonstrate the application of our proposed framework in identifying an appropriate response strategy, as illustrated in Figure 2.

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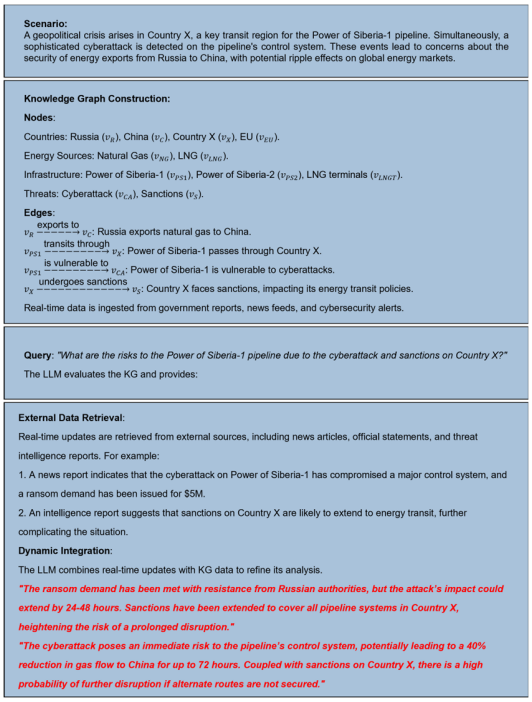


Figure 2. The example of our proposed framework.

Figure 3 shows the latency of pure LLM and the LLM with our proposed framework. In addition, Table 2 shows the average latency and token count of them.

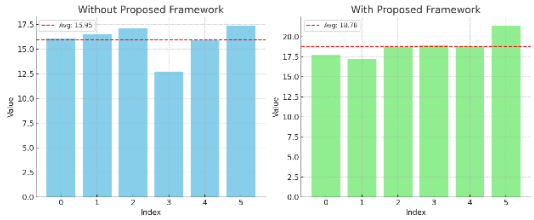


Figure 3. The latency of pure LLM and LLM with our proposed framework. The red line show the average latency.

Table 2. The average latency and token count of pure LLM and LLM with our proposed framework.

	Latency(s)	Token count
Without proposed framework	15.95	2049
With proposed framework	18.78	3642

V. Conclusions

This paper presents a novel framework for enhancing security and risk prevention in the critical Sino-Russian pipeline network. By integrating Knowledge Graph (KG) reasoning, Retrieval Augmented Generation (RAG), and large language models (LLMs), the approach enables real-time risk assessment, scenario planning, and proactive mitigation. The framework's comprehensive KG incorporates geopolitical, market, infrastructure, and cybersecurity data, allowing LLMs to reason dynamically via RAG. Security features like role-based access control and multi-factor authentication ensure system integrity. Simulated experiments highlight improved accuracy, timeliness, and effectiveness in risk identification compared to traditional methods. This adaptable solution advances critical infrastructure protection, enhancing energy corridor resilience and supporting a secure global energy future.

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References

[1] Skalamera, M. (2016). Sino-Russian energy relations reversed: A new little brother. Energy Strategy Reviews, 13, 97-108.

[2] Aggarwal, P. (2022). Sino-Russian energy relations: Opportunities and challenges. China's Energy Security: The Journey from Self-Sufficiency to Global Investor, 89-148.

[3] Wei, Q., Zhou, P., & Shi, X. (2024). Assessing the congestion cost of gas pipeline between China and Russia. Energy Strategy Reviews, 55, 101493.

[4] Korppoo, A., & Wang, Y. (2024). Sino–Russian Arctic Gas Cooperation Amid New Geopolitical Realities: Impacts and Perceptions. In China-Russia Relations in the Arctic: Friends in the Cold? (pp. 181-211). Cham: Springer Nature Switzerland.

[5] Liu, D. (2024). The politics of cross-border pipelines: Considering the resurgence of the Mongolian route for the Russia–China gas pipeline. Energy Research & Social Science, 113, 103551.

[6] Yao, Y., Duan, J., Xu, K., Cai, Y., Sun, Z., & Zhang, Y. (2024). A survey on large language model (llm) security and privacy: The good, the bad, and the ugly. High-Confidence Computing, 100211.

[7] Peng, C., Xia, F., Naseriparsa, M., & Osborne, F. (2023). Knowledge graphs: Opportunities and challenges. Artificial Intelligence Review, 56(11), 13071-13102.

[8] Fan, W., Ding, Y., Ning, L., Wang, S., Li, H., Yin, D., ... & Li, Q. (2024, August). A survey on rag meeting llms: Towards retrieval-augmented large language models. In Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (pp. 6491-6501).

[9] Cuconasu, F., Trappolini, G., Siciliano, F., Filice, S., Campagnano, C., Maarek, Y., ... & Silvestri, F. (2024, July). The power of noise: Redefining retrieval for rag systems. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval* (pp. 719-729).

[10] Arslan, M., Ghanem, H., Munawar, S., & Cruz, C. (2024). A Survey on RAG with LLMs. *Procedia Computer Science*, *246*, 3781-3790.

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